Worksheet 5-3 A Intervals of Increase and Decrease #1
AP Calculus Name_____

1. $x = \frac{\pi}{3}$ and $x = \frac{5\pi}{3}$

2. $x = \frac{7}{2}$

3. $x = e^{-1/2}$

4. $x=3, x=6, x=9, x=10$

$k'(x) = 0$ at $x=6$
$k'(x)$ is undefined at $x=3$, $x=9$
$x=10$

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5. g inc: $(-\infty, 0]$ g dec: $[0, +\infty)$

6. g inc: $[3, +\infty)$ g dec: $(-\infty, 3]$

7. g inc: never g dec: $(-\infty, -4) \cup (-4, +\infty)$

8. $m'(x)$ is increasing on $[-4, 0]$ and $[3, 10]$ because $m'(x) \geq 0$ on these intervals

9. $h(x)$ is inc: $(-\infty, 1]$ because $h'(x) > 0$
$h(x)$ is dec: $[1, +\infty)$ because $h'(x) < 0$